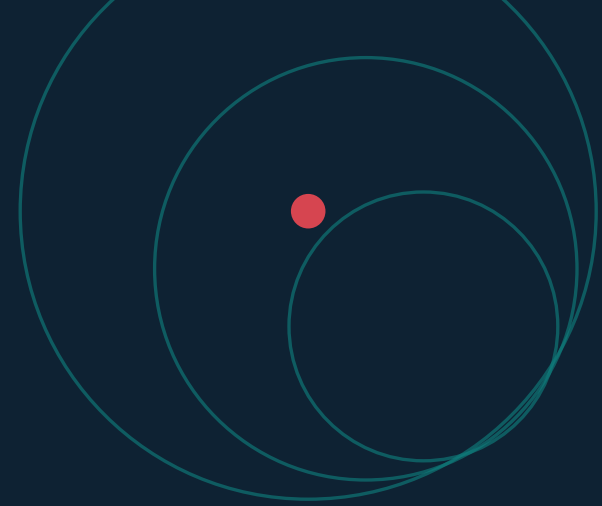


AI TRACK · IBM BOB HACKATHON

Drug Safety Signal Detector

Mining FDA adverse-event reports for statistically unusual drug-reaction patterns — before they become the next Vioxx.



TEAM

offset

TRACK

AI

TEAM LEAD

Mark Patel

MEMBERS

Mir Patel · Samad Sama · Keya Sonaiya

Dangerous patterns hide in plain sight

The FDA's adverse event database (FAERS) collects millions of “this drug caused a side effect” reports every year.

Pharmacovigilance teams can't manually scan that volume — so real safety signals can hide for years before anyone connects the dots.

- Who feels it: pharmacovigilance analysts, regulators, and ultimately every patient prescribed the drug.
- Why it matters: earlier detection means faster warnings, label changes, or withdrawals.

CASE IN POINT

Vioxx (rofecoxib)

27,000+

estimated excess heart attacks and cardiac deaths linked to the drug before it was voluntarily pulled from the market in 2004.

Early statistical signals existed in reporting data years before withdrawal — this is the gap our tool targets.

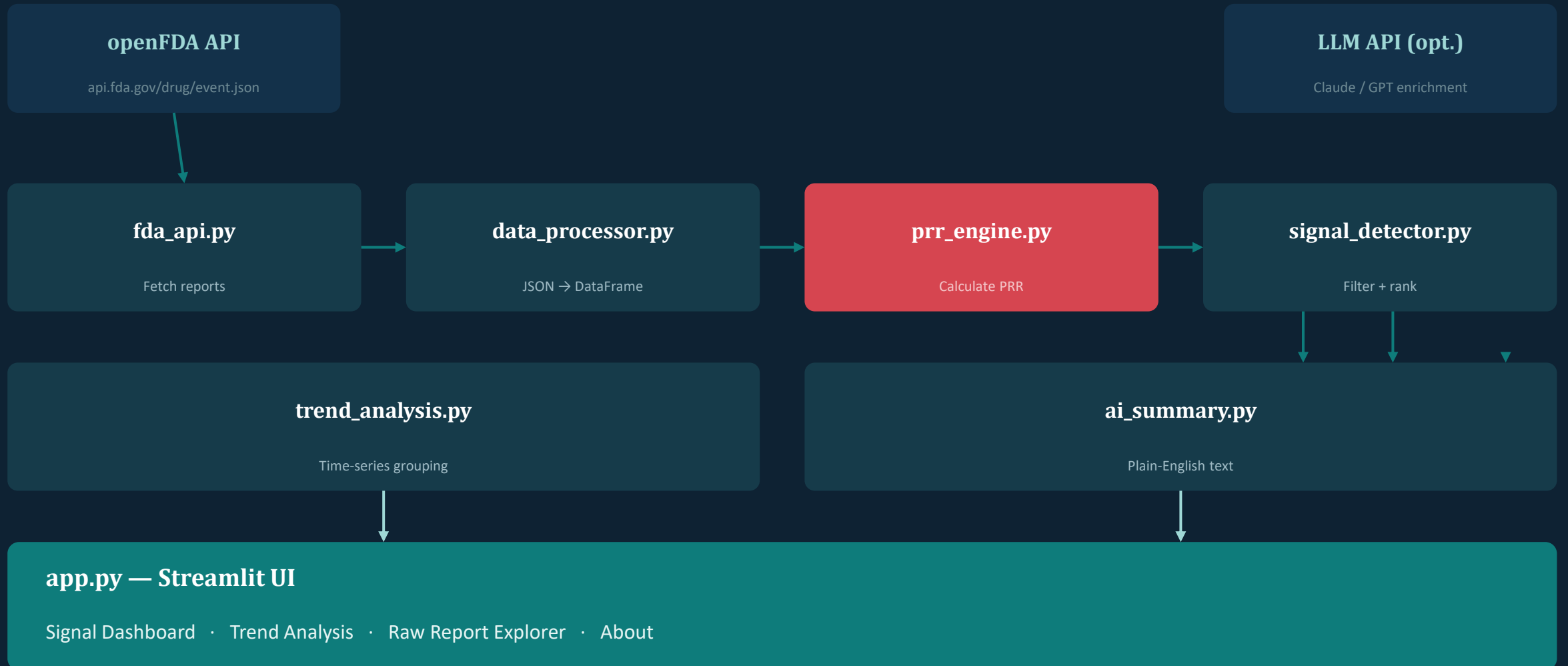
From raw FDA reports to ranked risk signals

We built a dashboard that pulls real openFDA adverse-event data, runs a proven statistical test on every drug–reaction pair, and surfaces the ones that look genuinely unusual — in plain English.



Every result is a statistical reporting signal, not a proven cause — the tool is explicit about that distinction throughout.

A single-process Streamlit app, six clear modules



The math behind the signal: PRR

Proportional Reporting Ratio

$$\text{PRR} = \left[\frac{A}{A+B} \right] \div \left[\frac{C}{C+D} \right]$$

- A** Reports with this drug AND this reaction
- B** Reports with this drug AND other reactions
- C** Reports with other drugs AND this reaction
- D** Reports with other drugs AND other reactions

Flag threshold: $\text{PRR} > 2$ (drug's reaction rate is more than double the background rate)

SIGNAL STRENGTH DASHBOARD

VIOXX

Myocardial infarction

3.2x

VIOXX

Cardiac arrest

2.6x

DRUG-B

Hepatic failure

2.1x

DRUG-C

Anaphylaxis

1.3x

"VIOXX shows a 3.2x higher rate of myocardial infarction than expected across all reported drugs."

Built with IBM Bob

IBM Bob was our AI development partner throughout the hackathon — not a runtime dependency, but the tool that shaped how fast and how cleanly we could build.

1

Scaffolding

Generated the initial module layout — `fda_api`, `data_processor`, `pr_engine`, `signal_detector`, `trend_analysis` — from our architecture sketch.

2

Statistical Review

Helped verify the PRR formula implementation $(A/(A+B) \div C/(C+D))$ against the standard pharmacovigilance definition.

3

Documentation

Co-authored architecture and setup docs so the repo is judge-runnable in minutes, no Docker required.

4

Iteration Speed

Rapid back-and-forth on the Streamlit dashboard let a 4-person team ship search, ranking, trends, and summaries in one hackathon window.

What success looks like

2.0x+

PRR threshold used to flag a candidate safety signal

4

dashboard views: signals, trends, raw reports, about

0

API keys required for the core detection workflow

Years → Minutes

potential head start on spotting a Vioxx-style pattern

WHAT WE'RE MOST PROUD OF

Real FDA adverse-event data paired with a transparent, explainable PRR method — turning large-scale reports into ranked signals while clearly separating statistical correlation from confirmed causation.

KNOWN LIMITATIONS

- Flags statistical signals — does not prove causation.
- Sensitive to reporting bias, duplicates, and missing data.
- Built to support pharmacovigilance analysis, not replace expert medical or regulatory review.

The people behind the signal



Mark Patel

TEAM LEAD

Project direction, PRR engine &
statistical validation



Mir Patel

MEMBER

openFDA integration & data
processing pipeline



Samad Sama

MEMBER

Streamlit dashboard, ranking & trend
visualization



Keya Sonaiya

MEMBER

Plain-English summary generation &
documentation